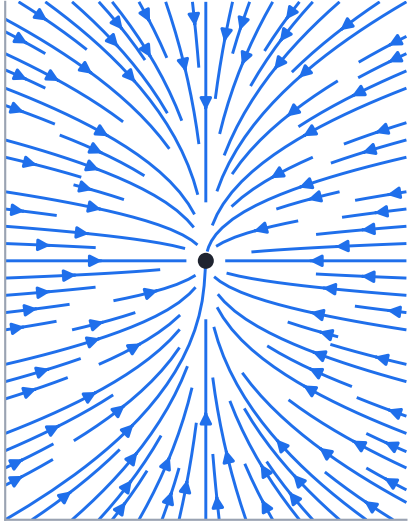
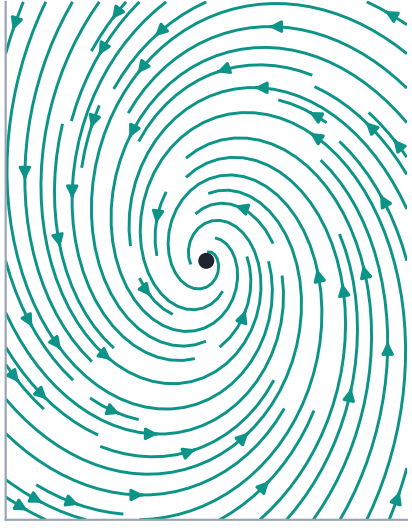
Linearise, look at the eigenvalues: the whole local theory

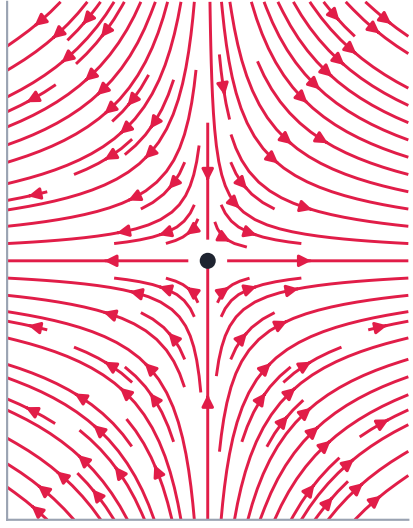
stable node
$\lambda_1, \lambda_2 < 0$



stable spiral
$\mathrm{Im}\,\lambda \neq 0$



saddle
$\lambda_1 > 0 > \lambda_2$



centre
$\mathrm{Re}\,\lambda = 0$

